

2. ANALYZING INCOME STATEMENTS

Expense Recognition

Three different methods are used to achieve this: the **matching principle**, **expensing as incurred**, and **capitalization**.

Matching principle, expenses to generate revenue are recognized in the same period as the revenue. Inventory provides a good example.

If purchased inventory unit > sales unit, unsold remaining move to Asset and become next year beginning inventory.

Capitalization is an application of the matching principle whereby costs are initially capitalized as assets on the balance sheet and then **expensed using depreciation or amortization**, to the income statement over the asset's life as its benefits are consumed.

Property, plant, and equipment (PP&E) and intangible assets with finite lives are examples of this

Capitalization include all costs of equipment that provide future economic benefits, including the costs that are necessary to get the asset **ready for use**.

Doesn't include training (people not equipment), repair (not initial cost), but include equipment rebuild to extend life (qualifies as a capital improvement)

Period costs, such as administrative costs, repair costs, are **expensed in the period incurred**.

Impact of capitalization on in financial statements: spread cost and tax over the period instead of huge cost in the first year. Net income is higher in Year 1 but lower in all the subsequent years

Capitalizing interest is added to **asset cost**; recognized via depreciation, increase asset in B/S (serve as the necessary cost for asset)

Research costs cannot be capitalized. However, **development costs** may be **capitalized**

NONRECURRING ITEMS

Unusual or infrequent items (**deducted from I/S before tax**):

- Gains or losses from the sale of assets or part of a business, if these activities are **not a firm's ordinary operations**

- Impairments, write-offs, and write-downs
- Restructuring costs

Discontinued operation (**deducted from I/S net of tax, before net income**):

One that management has decided to dispose of, but either has not yet done so, or has disposed of in the current year after the operation had generated income or losses

<i>Income Statement</i>
Net revenue
–COGS
Gross Profit
–SG&A
Operating Income
+/-Other gain or loss (Non operating item)
<i>+/-Unusual or infrequent items</i>
EBIT
–Interest expense
EBT
–Income tax expense
NI (Continuing operations)
<i>+/-Income from discontinued operation (net of tax)</i>
Net Income

EARNINGS PER SHARE

$$diluted EPS = \frac{Net\ Income - preferred\ dividends + Convertible\ dividends + Convertible\ debt\ interest}{time\ weighted\ outstandings + Convertible\ stocks + Treasury\ stocks}$$

Treasury Stocks shares =

$$\frac{average\ Px - option\ Px}{average\ PX} \times converted\ shares$$

If convertible is **anti-dilutive**, it should not be included in the diluted EPS calculation

The convertible are considered **anti-dilutive** when convertible EPS > basic EPS. Hence their inclusion would **increase the EPS**, rather than decrease it, which goes against the purpose of diluted EPS -> EPS thus = basic EPS

Shares Repurchased = Total Options * exercise price/market price

Treasury Stocks Shares = shares outstanding if exercise = Total Options -Shares Repurchased

If all preferred dividends are converted, preferred dividends = convertible dividends, leaving only Net Income in nominator.

Example: repurchase shares and issue 10% stock dividends

Shares outstanding on January 1:	$10,000 \times 1.10 \times 12/12$	of	=	11,000
the year				
Shares issued April 1:	$4,000 \times 1.10 \times 9/12$	of the year	=	3,300
Shares repurchased September 1:	$-3,000 \times 4/12$	of the year	=	<u>-1,000</u>
Weighted average shares outstanding			=	13,300

Check to see if diluted EPS < basic EPS. If the answer yes, the preferred stock is dilutive. If no, the preferred stock is antidilutive, and conversion effects are not included in diluted EPS.

Questions

30.1 revenue recognition

A contractor agrees to build a bridge for a total price of \$10 million. The project is expected to take four years to complete, at a total cost of \$6.5 million. After Year 1, costs of \$2.5 million have been incurred, and a further \$1 million in costs are incurred in Year 2. The client pays \$2 million in each of the first two years. The amount of revenue the contractor should recognize in Year 2 is closest to:

ans:

Year1: $2.5/6 =$ cost percentage. year revenue = Total revenue x percentage = 3.85

Year2: $3.5/6 =$ cost percentage. year revenue = Total revenue x percentage - Year1 revenue = $5.38 - 3.85 = 1.47$

The amount paid by the client does not affect revenue. Only cost ratio is considered.

A realtor sells one of its client's houses for \$1.4 million, earning a commission of 3%. The costs to the realtor associated with the sale are \$15,000. What is the realtor's gross profit for this transaction?

ans:

gross profit = $1.4 \times$ commission - costs of sales = 27,000

30.2 expense recognition

The cost of an intangible asset is most likely to be amortized if the asset has a(n):

ans:

finite life and was purchased.

Intangible assets with an **infinite life** are **not amortized** because they are expected to provide economic benefits to the company indefinitely. These assets are subjected to **annual impairment testing** to ensure they are not overvalued on the balance sheet.

30.3 nonrecurring items

Changing an accounting estimate:

ans:

is reported prospectively. No restatement of prior-period statements is necessary.

Need retrospective change (prior period change):

1. Change in accounting policy
2. A correction of an accounting error made

30.4 earnings per share

An analyst has gathered the following information about a company:

- 300,000 shares outstanding
- 100,000 warrants exercisable at \$50 per share
- Average share price: \$55
- Year-end share price: \$60

How many shares should be used in computing diluted EPS?

ans:

Because the exercise price of the warrants is < the average share price, the warrants are dilutive (exercisable by holders, increase shares outstanding after execution). Use the Treasury stock method:

$55 - 50 / 55 \times 100,000 = 9091$ shares

An analyst has gathered the following information about a company:

- 100,000 common shares outstanding from the beginning of the year
- Earnings: \$125,000
- 1,000, 7%, \$1,000 par bonds convertible into 25 shares each, outstanding as of the beginning of the year
- Tax rate: 40%

The company's diluted EPS is closest to?

ans:

Basic EPS = $125,000 / 100,000 = 1.25$

dilutive EPS:

Adjusted Earnings = $125K + 1K \times 1K \times 7\% \times (1 - 0.4) = 125K + 42K = 167K$

shares outstandings = $100K + 25K = 125K$

dilutive EPS = 1.34

dilutive EPS > basic EPS, it is anti dilutive, remain EPS = 1.25

An analyst has gathered the following information about a company:

- 50,000 common shares outstanding from the beginning of the year
- Warrants outstanding all year on 50,000 shares, exercisable at \$20 per share
- Stock is selling at year-end for \$25
- Average price of the company's stock for the year was \$15

How many shares should be used in calculating the company's diluted EPS?

ans:

The warrants are antidilutive. The average price per share of \$15 is less than the exercise price of \$20. The year-end price per share is not relevant. The denominator consists of only the common stock 50K for basic EPS.